

Day 3 — Neem — Bitter Healer

Module: Herbs — Auśadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify nimba; learn about azadirachtin; understand high-dose safety warning.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *nimba* (IAST: nimba; lit. “neem — *Azadirachta indica*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Botanical identification.** *Azadirachta indica*. Tall tree, ~15-20 m. Pinnate compound leaves with serrated leaflets. Distinctive bitter smell when crushed.
2. **Active compound.** Azadirachtin (in seeds, oil). Nimbidin (in leaves, bark). Quercetin (a flavonoid).
3. **Traditional uses.** Caraka and Suśruta cite *nimba* for: skin conditions, fever, infections, oral hygiene (chewing the twig). External use is well-supported by modern studies.
4. **THE SAFETY WARNING.** Neem at high concentrations (especially neem oil) is documented to be hepatotoxic. Caraka itself notes restrictions. Modern pediatric case-reports (e.g. *Indian Pediatrics* journals) document harm in infants given neem oil. NEVER ingest in concentrated form. State this rule THREE TIMES.
5. **Modern use cases.** Neem-based pesticides (azadirachtin is one of the most studied bio-pesticides). Toothpaste with neem extract. Skin creams. Discussion: “*What separates safe external use from risky internal use?*”

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Neem — Bitter Healer.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).